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[22MSCH3H03]

THIRD SEMESTER UNIVERSITY EXAMINATION, JANUARY - 2025

PHARMACIEUTICAL AND INDUSTRIAL CHEMISTRY

Time: 03 HOURS **Maximum Marks:50**

		Blooms Level	CO	Marks
	SECTION – A			
1.	**Answer any FOUR of the following questions.**			**4×5=20**
a.	What is a drug? Explain the qualities of a drug.	4	1	**05**
b.	What are analgesic and antipyretic drug? Explain the classification of analgesic and antipyretics drug with examples.	2		**05**
c.	Write the structures and application of allethrin and cypermethrin.		4	**05**
d.	Explain the role of Zeigler-Natta catalyst in polymerization reactions and its limitations.	4	5	**05**
e.	Describe characteristics of the fractional distillation process.	2	6	**05**
	SECTION – B			
2.	**Answer any TWO of the following questions.**			**2×9=18**
a.	Describe optimization of drug-target interaction on the basis of variation of substituents, ring variation, isosteres and bioisosteres, simplification of the structure, and rigidification of the structure.	4	1	**09**
b.	Explain the terms agonist, partial agonist, intrinsic activity, antagonist, occupation, rate, and induced fit theories.	4	2	**09**
c.	**i.** Explain the synthesis of ibuprofen from isobutyl benzene. **ii.** Write a note on synthesis of sorbitrate and methyldopa.			**5+4**
	SECTION – C			
3.	**Answer the following question.**			**1x12=12**
a.	**i.** Write a note on characteristics of pyrethroids.	4	4	**04**
	ii. What are polyamides? Write a note on nylons.	4	5	**04**
	iii. Classify petroleum based on chemical composition, density, and viscosity.	3	6	**04**